2026/ICT450

UNIVERSITI TEKNOLOGI
MARA FINAL EXAMINATION

COURSE	DATABASE DESIGN AND DEVELOPMENT
COURSE CODE	ICT450
EXAMINATION	MOCK 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of five (5) questions.

2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.

3. Do not bring any material into the examination room unless permission is given by the invigilator.

4. Please check to make sure that this examination pack

   consists of: i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty

5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

QUESTION 1

Nusantara Learning Hub currently keeps student records in separate spreadsheet files. The manager is considering a database management system (DBMS).

a) Define database, DBMS and database system.

(6 marks)

b) State FOUR limitations of the file-based approach that a DBMS addresses.

(8 marks)

c) Explain THREE functions of a DBMS that address those limitations.

(6 marks)

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QUESTION 2

The relation schema ENROL(StudentID, CourseID, StudentName, CourseName, Lecturer, Grade) records student enrolment. Assume StudentID and CourseID together uniquely identify a row.

a) Identify the primary key and TWO candidate keys.

(4 marks)

b) Explain entity integrity and referential integrity with one example each.

(6 marks)

c) Define tuple, attribute, degree and cardinality.

(4 marks)

d) Explain TWO update anomalies that can occur in this schema.

(6 marks)

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QUESTION 3

Cedar Community Library lends books to members. Each book has one publisher; a publisher can publish many books. A member can borrow many books over time, and each loan records the loan date and due date.

a) Identify the entities, relationships and cardinalities.

(8 marks)

b) Draw an entity-relationship diagram for the scenario.

(6 marks)

c) Map the ERD to a relational schema and mark primary and foreign keys.

(6 marks)

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QUESTION 4

The relation ORDER(OrderID, CustomerID, CustomerName, ProductID, ProductName, Quantity, UnitPrice) has these functional dependencies: OrderID → CustomerID, CustomerName; CustomerID → CustomerName; ProductID → ProductName; (OrderID, ProductID) → Quantity, UnitPrice.

a) Identify the candidate key and all functional dependencies.

(4 marks)

b) Explain 1NF, 2NF and 3NF.

(6 marks)

c) Identify the partial and transitive dependencies.

(4 marks)

d) Decompose the relation into 3NF relations.

(6 marks)

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QUESTION 5

Sri Murni Health Clinic is planning a database for patients, doctors, appointments and prescriptions. The project will follow the database life cycle.

a) List FOUR phases of the database life cycle and state one activity in each.

(8 marks)

b) Differentiate the SDLC and the DBLC.

(4 marks)

c) Compare top-down and bottom-up database design.

(4 marks)

d) Explain centralized and decentralized design.

(4 marks)

END OF QUESTION PAPER

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